State-Dependent Resource Allocation Under Uncertainty: A Mixed-Integer Stochastic Programming Approach

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Abstract

This paper develops a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model addresses the fundamental challenge of optimally allocating limited time and cognitive resources across competing task categories when system state evolves dynamically and recovery requirements are uncertain. We demonstrate that optimal allocation strategies exhibit strong state-dependency, with the correlation between initial completion state and priority task allocation reaching $r = -0.91$. Computational experiments across six scenarios reveal that early-stage decisions have outsized importance, with sensitivity coefficients of $\partial z^*/\partial g_0 = -709$ at low completion states diminishing to near-zero at higher completion levels. The model achieves optimal solutions within 60 seconds using commercial MILP solvers, making it suitable for practical decision support applications. These findings contribute to the operations research literature on resource allocation and provide actionable insights for personal productivity management systems.

Keywords: Resource Allocation, Mixed-Integer Programming, Stochastic Optimization, Decision Support Systems, Rolling Horizon Planning

1. Introduction

The efficient allocation of limited resources across competing activities represents a fundamental challenge in operations research and management science. As the modern workplace increasingly demands that individuals manage multiple professional and personal obligations under cognitive and temporal constraints, sustainable life management has emerged as an urgent research priority. Recent evidence suggests that billions of working days are lost annually due to mental health issues stemming from cognitive overload and excessive job demands [1]. The prevalence of burnout continues to rise across sectors, with physicians spending over five hours daily on electronic health record systems and additional time after work hours, contributing to physician attrition and appointment delays [2]. These trends underscore the critical importance of developing systematic approaches to resource allocation that account for human cognitive limitations and well-being.

Traditional time management strategies often fail because they ignore the finite nature of human lifespan and cognitive capacity [3]. Productivity depends more fundamentally on managing attention than on rigid time scheduling, as cognitive limitations make it difficult to sustain focus across competing demands in dynamic environments [4]. The increasing reliance on artificial intelligence and algorithmic management systems introduces additional complexity: while AI promises efficiency gains in documentation and task automation, it also risks depleting human cognitive resources and fostering decision-making that overlooks emotional and contextual nuances [5]. Forecasts suggest near-total AI dominance in key domains by 2027, potentially reshaping how humans allocate limited time and cognitive effort in work systems [6]. Human resource management professionals themselves are overworked with increasingly complex jobs, making many suffer from burnout and leaving insufficient time for strategic issues [7].

The urgency of this research agenda is amplified by the growing recognition that health status itself constitutes a resource. Studies demonstrate that chronic illness symptom severity is positively associated with burnout and negatively associated with work engagement, even after controlling for job demands and resources [8]. With over half the current workforce managing one or more chronic conditions, workplace interventions that improve employee health also enhance perceived organizational support, reduce burnout, and increase work engagement [9]. The COVID-19 pandemic has further highlighted these challenges, as maintaining traditional approaches to work during crises predisposes systems to diminished quality of care and increased burnout risk [10]. Research on STEM women recovering from burnout reveals that marginalized workers in highly institutionalized settings face unique challenges in the aftermath of burnout, with important implications for workforce sustainability [11].

The gig economy introduces distinct resource allocation challenges through algorithmic management. Research shows that algorithmic evaluation and discipline increase job burnout by negatively impacting workers' basic psychological needs, while algorithmic direction can alleviate burnout by enhancing autonomy and competence [12]. This dual nature of technology-mediated work extends to augmentation-based AI tools: while frequent usage leads to knowledge gain and improved task performance, it also causes information overload that impairs both performance and recovery [13]. Technostressors stemming from information technology use have become a major source of stress, negatively affecting work-life balance through emotional exhaustion, though job self-efficacy can buffer these effects [14]. Understanding how configurations of technostressors interact to produce burnout represents a critical research frontier [15].

The work-family interface presents another dimension of the sustainable life management challenge. Work-family conflict research has predominantly used composite-level approaches, but dimension-level analysis reveals that time-based, strain-based, and behavior-based conflicts operate through distinct mechanisms [16]. Emotional resource accumulation, distinct from emotional resource depletion, helps explain how work demands affect family functioning and employee well-being [17]. Flexible working arrangements offer one potential solution: meta-analytic evidence shows that having both schedule and location flexibility is more beneficial than either alone, and that flexibility availability may be more important than actual use [18]. Experimental evidence confirms that smart working increases productivity while improving work-life balance, with men also increasing time dedicated to household activities [19]. Compressed workweek implementations show that work-life balance increases while fatigue and time pressure decrease, with effects contingent on employee expectations [20].

Resource theory provides essential foundations for understanding these dynamics. Conservation of Resources (COR) theory and the Job Demands-Resources (JD-R) framework explain how job demands and resources relate to occupational strain, with job resources moderating the demand-strain relationship [21]. Research extends JD-R theory beyond the work domain to show how senior executives utilize both internal and external resources for coping and resource creation [22]. Supervisor resilience can cross over to employees, promoting well-being through lower burnout, lower psychological distress, and greater life satisfaction [23]. Managers can help employees navigate psychologically difficult decisions through three forms of self-management: equipping, conserving, and restoring the self across preparation, execution, and transition phases [24].

Optimization and decision support approaches offer promising avenues for addressing these challenges. Constraint programming models for personal process management demonstrate effectiveness in reducing cognitive load and improving planning outcomes in dynamically changing situations [25]. Just-in-time access to knowledge repositories reduces cognitive load, enables learning, and builds problem-solving capabilities [26]. In healthcare, routing and scheduling optimization can improve continuity of care while enhancing resource utilization [27]. Workforce sizing and shift scheduling decisions can balance efficiency with shift stability to improve worker well-being without sacrificing profitability [28]. Two-stage stochastic game models can guide capacity planning investments to address surgical waiting lists while accounting for uncertainty [29]. Managing flexible capacity in service systems helps prevent worker burnout in understaffed systems while improving service quality [30].

This paper develops a mixed-integer stochastic programming formulation that captures these dynamics through explicit state tracking and uncertainty modeling. The proposed model makes three primary contributions to the literature. First, we formulate the multi-period resource allocation problem as a tractable MILP with state-dependent objective weights, enabling efficient solution using commercial solvers. Second, we incorporate uncertainty in recovery requirements through a scenario-based stochastic programming approach, providing robustness against variable cognitive demands. Third, we provide comprehensive computational experiments demonstrating the sensitivity of optimal allocations to initial conditions, with important implications for decision support system design.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on resource allocation, cognitive load theory, and work-life sustainability. Section 3 presents the mathematical formulation, including decision variables, constraints, and the objective function. Section 4 describes the computational experiments and presents the main empirical findings. Section 5 discusses the managerial implications of the results, and Section 6 concludes with directions for future research.

2. Literature Review

This section situates our work within five interconnected streams of research: (1) cognitive load and attention management, (2) job demands, resources, and burnout, (3) work-life balance and flexibility, (4) technology and AI in the workplace, and (5) optimization approaches to resource allocation.

2.1. Cognitive Load and Attention Management

Cognitive load theory provides foundational insights for sustainable life management. Research demonstrates that cognitive training yields lasting benefits, with reasoning and speed-of-processing training showing effects up to 10 years in older adults, supporting strategies for sustaining cognitive resources across the lifespan [31]. Contemporary applications of cognitive load theory in computer-based environments reveal the importance of worked examples, split-attention effects, and the self-management of cognitive load [32]. NSF-funded research on peak productivity develops tools using smartphones and smartwatches to track biological clocks and align tasks with peak alertness periods, addressing disruptions from mismatched timing that cause health issues and errors [33].

The distinction between time management and attention management is critical. Traditional time management strategies fail because they ignore cognitive constraints, forcing individuals to confront limitations and make hard choices among competing priorities rather than attempting to optimize everything [3]. Productivity hinges more on managing attention than rigid time scheduling, as cognitive limitations make it difficult to sustain focus across competing demands [4]. This recognition has spurred interest in decision support systems that account for cognitive constraints. Research shows that just-in-time access to codified knowledge reduces cognitive load and enables learning, with workers accessing procedural knowledge from document repositories building firm-specific human capital [26].

Human-AI collaboration offers promise for reducing cognitive burden but requires careful design. Studies find that granting users decision control improves trust and compliance with AI recommendations, while explanations may increase task complexity depending on the user's cognitive ability [34]. The challenge is particularly acute for knowledge workers facing information overload. Mitigation strategies include streamlining information, investing in training and mental health resources, promoting deep work with breaks, and ensuring equitable workload distribution [1].

2.2. Job Demands, Resources, and Burnout

The Job Demands-Resources (JD-R) model provides a powerful framework for understanding sustainable life management. Research confirms that job resources moderate the relationship between job demands and occupational strain, with autonomy reducing strain under high workload conditions [21]. Extensions of JD-R theory beyond the work domain reveal that senior executives utilize both internal and external resources, and that job crafting creates adaptable resources to meet evolving demands [22]. Individual health status itself functions as a resource: chronic illness symptom severity predicts burnout and reduced work engagement above and beyond the effects of job demands and resources [8].

Conservation of Resources (COR) theory complements JD-R by explaining resource dynamics. Research distinguishes emotional resource accumulation from depletion, developing measurement instruments that capture how resource-generating activities (work engagement, co-worker support) and resource-depleting activities (job demands, emotional labor) differentially predict family functioning and life satisfaction [17]. Supervisor resilience can cross over to employees, with time-lagged studies demonstrating that supervisor resilience promotes employee well-being through lower burnout and psychological distress [23]. Managers can help employees navigate resource-intensive tasks through equipping, conserving, and restoring the self across task phases [24].

Burnout represents a critical outcome when resource allocation fails. Healthcare providers face particular challenges: physicians' workflow decisions regarding when to perform electronic health record tasks significantly influence total workload and after-hours work, with implications for sustainable practice [2]. STEM women recovering from burnout employ distinct sensemaking strategies (combative, regenerative, and promissory) to renegotiate work trajectories, revealing the relational negotiation of protecting, investing, and fostering resources during disruption [11]. Workplace interventions like the Live Healthy, Work Healthy program improve perceived organizational support, reduce burnout, and increase work engagement through structured support for chronic condition management [9].

2.3. Work-Life Balance and Flexibility

Work-family conflict research has evolved from composite-level to dimension-level analysis. Meta-analytic evidence demonstrates that time-based, strain-based, and behavior-based dimensions of conflict have distinct theoretical underpinnings and empirical correlates, with resource allocation theory particularly relevant to time-based conflict [16]. Crowdsourced firm ratings analysis reveals that work-life balance ratings influence organizational outcomes differently across industries, suggesting that decision makers should tailor motivation strategies to workforce needs [35].

Flexible working arrangements (FWAs) offer promising solutions to work-life challenges. Comprehensive meta-analysis identifies antecedents of FWA availability and use (managerial status, self-efficacy, task interdependency) and confirms beneficial outcomes including job satisfaction, organizational commitment, and family satisfaction [18]. Importantly, having both flextime and flexplace proves more beneficial than either alone, and availability may matter more than actual use. Experimental evidence from a randomized trial demonstrates that flexible place and time of work increases productivity while improving work-life balance, with men also increasing household and care time [19].

Alternative work schedules provide additional flexibility mechanisms. Longitudinal research on compressed workweek implementation shows that work-life balance increases while fatigue and time pressure decrease, with effects contingent on employee expectations regarding the schedule change [20]. The future of work increasingly involves hybrid arrangements despite return-to-office pressures, with research showing that work-from-anywhere policies can enhance performance through flexibility [36].

2.4. Technology and AI in the Workplace

Technology plays a dual role in sustainable life management, serving as both enabler and stressor. Technostressors from information technology use have become a major stress source, negatively affecting work-life balance through emotional exhaustion, though job self-efficacy buffers these effects [14]. Configurational analysis reveals that interdependencies among technostressors (complementarity, contingency, and substitution) form patterns leading to burnout and reduced job performance, explaining why interventions addressing independent technostressors may fail [15]. Information and communication technology can reduce nurses' burden by facilitating environmental management and non-contact communication while providing emotional support for patients, though implementation requires attention to digital health literacy and safety [10].

Artificial intelligence introduces new dynamics into resource allocation. Reliance on algorithmic management in healthcare risks depleting human cognitive resources by favoring data summaries over nuanced listening, fostering cognitive miserliness and deskilling [5]. In the gig economy, algorithmic evaluation and discipline increase burnout by negatively impacting psychological needs, while algorithmic direction can alleviate burnout by enhancing autonomy [12]. Research on augmentation-based AI reveals countervailing mechanisms: frequent usage leads to knowledge gain and improved performance, but also causes information overload impairing performance and recovery [13]. Generative AI can serve as a helpful assistant for both strategic and operational tasks when implemented with appropriate prompts and verification processes [7]. Predictions of near-total AI dominance in key domains by 2027 raise fundamental questions about how humans will allocate limited time and cognitive effort as work systems evolve [6].

Human-robot collaboration in manufacturing demonstrates the potential for dynamic task allocation. Research on recyclofacturing emphasizes dynamic task distribution between humans and robots in assembly and welding, with attention to human factors for safe collaboration [37]. The key insight is that technology should adapt task distribution to evolving conditions and recovery needs in human-machine teams.

2.5. *Optimization Approaches to Resource Allocation*

Operations research provides powerful tools for sustainable life management. Constraint programming models for personal process management decrease cognitive load from decision-making and generate positive user experiences, enabling better planning in less time and fast replanning in dynamically changing situations [25]. Behavioral research demonstrates that simpler yet efficient contracts can mitigate the trade-off between efficiency and complexity, explaining the prevalence of minimum order quantity terms in supply contracts [38]. Understanding incentive structures is critical: research shows that incentivizing solely estimation accuracy can result in hidden inefficiency from strategic schedule inflation, while combined accuracy and speed incentives yield compressed but reliable schedules [39].

Healthcare resource allocation presents particularly challenging optimization problems. First Route Second Assign decomposition approaches enforce continuity of care in home health care while improving routing efficiency [27]. Two-stage stochastic game models guide capacity planning investments to address surgical waiting lists while accounting for demand uncertainty [29]. Multistage staffing strategies utilizing capacities of various flexibility help manage service systems facing worker shortages, preventing burnout in understaffed systems while improving service quality [30].

Workforce sizing and scheduling optimization can balance multiple objectives. Research on last-mile delivery demonstrates that introducing limited shift flexibility achieves similar effects as completely flexible systems while ensuring better work-life balance for workers, showing that improved working conditions need not sacrifice profitability [28]. This insight that stability and efficiency can coexist underlies our approach to sustainable life management.

2.6. *Research Gap and Contribution*

Despite the extensive literature on individual aspects of cognitive load, burnout, work-life balance, technology, and optimization, no existing framework integrates these streams into a unified optimization model for personal resource allocation. Prior work either focuses on organizational interventions without mathematical formalization or develops optimization models without accounting for cognitive constraints and recovery dynamics. Our contribution addresses this gap by developing a tractable mixed-integer stochastic programming formulation that captures state-dependent prioritization, uncertain recovery requirements, and multi-dimensional value creation, providing a foundation for decision support systems that promote sustainable life management.

3. Model Formulation

This section presents the mathematical formulation of the multi-period resource allocation problem. We begin by defining the fundamental sets and indices, then specify the decision variables, parameters, constraints, and objective function.

3.1. Problem Setting

Consider a decision-maker who must allocate available time across $|\mathcal{I}|$ task categories over a planning horizon of T days. Each task category $i \in \mathcal{I}$ is characterized by its resource consumption rates, minimum and maximum time bounds, and contribution to multiple value dimensions. The decision-maker faces uncertainty regarding recovery requirements following periods of high cognitive load, which we model through a discrete set of scenarios $\mathcal{S}$ with associated probabilities.

The problem exhibits two key structural features that distinguish it from standard resource allocation formulations. First, a critical priority task (Task 1) has a well-defined completion target, and progress toward this target affects both the objective function weights and the resource constraints. Second, cognitive recovery following emotionally demanding work requires uncertain amounts of time, reducing availability for productive activities.

3.2. Notation Reference

Table 1 provides a comprehensive reference for all mathematical notation used throughout this paper.

3.3. Sets and Indices

We define the fundamental index sets that structure the mathematical formulation. The planning horizon is represented by $\mathcal{T} = \{1, 2, \ldots, T\}$, where T denotes the planning cycle length (e.g., $T = 30$ for monthly planning). The set of task categories $\mathcal{I} = \{1, 2, \ldots, |\mathcal{I}|\}$ encompasses all activities competing for the decision-maker's limited resources. For presentation purposes, each task is anonymized: Task 1 represents a high-priority time-critical activity, Task 2 corresponds to career development, Task 3 captures scheduled obligations, Task 4 represents recovery and restoration, Task 5 denotes long-term capital building activities, Task 6 captures social engagement, and Task 7 represents physical maintenance activities.

Uncertainty in recovery requirements is modeled through the discrete scenario set $\mathcal{S} = \{1, 2, \ldots, |\mathcal{S}|\}$, where each scenario s specifies a required recovery duration β_s with probability p_s. Based on empirical calibration, we set $\beta_s \in \{1, 2, 3, 4\}$ hours with corresponding probabilities $p_s = \{0.4, 0.3, 0.2, 0.1\}$ in our computational experiments.

3.4. Decision Variables

For each task $i \in \mathcal{I}$ and day $t \in \mathcal{T}$, we define two sets of primary decision variables. The binary variable $x_{it} \in \{0, 1\}$ equals 1 if task i is undertaken on day t and 0 otherwise, while the continuous variable $h_{it} \in \mathbb{R}_+$ specifies the hours allocated to task i on day t. These variables are linked through logical constraints ensuring that positive time allocation requires task selection.

State variables track progress on the priority task over time. Let $G_t \in \mathbb{R}_+$ denote the cumulative hours of Task 1 work completed by the end of day t, and define $g_t = G_t / G^{\text{total}} \in [0, 1]$ as the proportion of total required work completed. The security level $\sigma_t \in [0, 1]$, derived from g_t, captures how progress on the priority task affects available resource capacity.

For uncertainty modeling, we introduce scenario-indexed auxiliary variables. The continuous variable $r_{ts} \in \mathbb{R}_+$ represents recovery hours required on day t under scenario s, while the binary indicator $z_{ts} \in \{0, 1\}$ flags days with high emotional load under scenario s.

Table 1: Complete Notation Reference

Symbol	Description
Sets and Indices	
$\mathcal{T}$	Set of days in planning horizon, $\{1, 2, \ldots, T\}$
$\mathcal{I}$	Set of task categories
$\mathcal{S}$	Set of recovery scenarios
T	Planning horizon length (days)
i	Index for tasks, $i \in \mathcal{I}$
t	Index for days, $t \in \mathcal{T}$
s	Index for scenarios, $s \in \mathcal{S}$
Decision Variables	
x_{it}	Binary indicator: 1 if task i undertaken on day t
h_{it}	Continuous hours allocated to task i on day t
G_t	Cumulative hours of Task 1 completed by end of day t
g_t	Completion proportion for Task 1, $g_t = G_t / G^{\text{total}}$
σ_t	Security level derived from g_t
r_{ts}	Recovery hours required on day t under scenario s
z_{ts}	Binary indicator for high emotional load on day t, scenario s
Resource Parameters	
$H^{\max}$	Maximum usable hours per day
$E^{\max}$	Maximum energy units per day
$M^{\max}$	Maximum emotional capacity units per day
e_i	Energy consumption rate for task i (units/hour)
m_i	Emotional cost rate for task i (units/hour)
$h_i^{\min}$	Minimum hours required if task i is selected
$h_i^{\max}$	Maximum hours allowed for task i
G^{total}	Total required hours for Task 1 completion
Return Parameters	
α_i^I	Priority task return rate for task i
α_i^Y	Income/employability return rate for task i
α_i^K	Long-term capital return rate for task i
α_i^O	Optionality return rate for task i
Weight Parameters	
$w_I(t, g_t)$	Time and state-dependent priority task weight
$w_Y(t), w_K(t), w_O(t)$	Time-dependent weights for other dimensions
$w_I^0, w_Y^0, w_K^0, w_O^0$	Base weights for each return dimension
$\rho_I, \rho_Y, \rho_K, \rho_O$	Growth rates for weight evolution
λ_1	Cost coefficient for energy consumption
$\lambda_2(g_t)$	State-dependent cost coefficient for emotional load
δ	Daily discount factor
Stochastic Parameters	
β_s	Required recovery hours under scenario s
p_s	Probability of scenario s
M	Big-M constant for logical constraints

3.5. Parameters

Table 2 presents the task-specific parameters used in the computational experiments. These values were calibrated based on typical resource consumption patterns and represent averages across task instances.

Table 2: Task Parameter Values

Task	Energy (units/hr)	Emotion (units/hr)	Min Hours	Max Hours
Task 1	15	12	1	6
Task 2	12	10	1	4
Task 3	10	5	2	8
Task 4	0	-5	1	4
Task 5	12	8	1	6
Task 6	5	-3	0	3
Task 7	8	-4	0	2

The daily resource budgets are set at $H^{\max} = 16$ usable hours, $E^{\max} = 100$ energy units, and $M^{\max} = 100$ emotional capacity units. These values reflect physiological constraints on sustained cognitive effort.

3.6. Objective Function

The objective function maximizes expected discounted utility over the planning horizon, accounting for multiple value dimensions and resource costs. Formally, we solve:

$$\max \quad \mathbb{E}_s \left[\sum_{t=1}^{T} \delta^t \sum_{i \in \mathcal{I}} U_{it}(h_{it}, g_t) \right] \tag{1}$$

where $\delta = 0.95$ is the daily discount factor and the utility function is defined as:

$$\begin{aligned} U_{it}(h_{it}, g_t) = {} & w_I(t, g_t) \cdot \alpha_i^I \cdot h_{it} + w_Y(t) \cdot \alpha_i^Y \cdot h_{it} \\ & + w_K(t) \cdot \alpha_i^K \cdot h_{it} + w_O(t) \cdot \alpha_i^O \cdot h_{it} \\ & - \lambda_1 \cdot e_i \cdot h_{it} - \lambda_2(g_t) \cdot m_i \cdot h_{it} \end{aligned} \tag{2}$$

The four return dimensions correspond to priority task progress (α_i^I), income and employability gains (α_i^Y), long-term capital accumulation (α_i^K), and optionality improvement (α_i^O). The weights on these dimensions evolve dynamically based on time and state according to:

$$w_I(t, g_t) = w_I^0 \cdot (1 - g_t) \cdot \exp(-\rho_I t/T) \tag{3}$$

$$w_Y(t) = w_Y^0 \cdot (1 + \rho_Y t/T) \tag{4}$$

$$w_K(t) = w_K^0 \cdot (1 + \rho_K t/T) \tag{5}$$

$$w_O(t) = w_O^0 \cdot (1 + \rho_O t/T) \tag{6}$$

This structure captures the intuition that priority task urgency decreases as completion progresses ($1 - g_t$ factor) and as time passes (exponential decay), while longer-term objectives gain relative importance over the planning horizon.

3.7. Constraints

The model includes several constraint families ensuring feasibility and logical consistency.

Time availability. Total time allocation plus required recovery cannot exceed daily available hours:

$$\sum_{i \in \mathcal{I}} h_{it} + r_{ts} \leq H^{\max} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \tag{7}$$

Logical linking. Time allocation is bounded by task-specific minimums and maximums when the task is selected:

$$h_i^{\min} \cdot x_{it} \leq h_{it} \leq h_i^{\max} \cdot x_{it} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{8}$$

Resource consumption. Energy and emotional load cannot exceed state-adjusted capacity:

$$\sum_{i \in \mathcal{I}} e_i \cdot h_{it} \leq E^{\max} \cdot (1 + 0.2 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \tag{9}$$

$$\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \leq M^{\max} \cdot (1 + 0.3 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \tag{10}$$

Stochastic recovery. Days with high emotional load trigger scenario-dependent recovery requirements:

$$\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \geq 0.8 \cdot M^{\max} - M \cdot (1 - z_{ts}) \quad \forall t, s \tag{11}$$

$$r_{ts} \geq \beta_s \cdot z_{ts} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \tag{12}$$

State evolution. Priority task progress accumulates over time:

$$G_t = G_{t-1} + h_{1,t} \quad \forall t \in \mathcal{T} \tag{13}$$

$$g_t = \min\left(1, \frac{G_t}{G^{\text{total}}}\right) \quad \forall t \in \mathcal{T} \tag{14}$$

3.8. Model Properties

The formulation constitutes a mixed-integer linear program with $O(|\mathcal{I}| \cdot T \cdot |\mathcal{S}|)$ binary and continuous variables and constraints. For typical parameter values (e.g., $|\mathcal{I}| = 7$, $T = 30$, $|\mathcal{S}| = 4$), this yields approximately 840 binary variables, 840 continuous variables, and 1,680 constraints. This problem size is well within the capability of modern MILP solvers such as Gurobi and CPLEX.

Proposition 1 (Feasibility)**.** *The model is feasible if and only if* $\sum_{i \in \mathcal{I}} h_i^{\min} + \max_s \beta_s \leq H^{\max}$.

The proposition follows directly from observing that the binding constraints involve satisfying all minimum task requirements while accommodating the worst-case recovery scenario.

4. Computational Experiments

This section presents the computational experiments designed to investigate the sensitivity of optimal resource allocations to initial conditions and planning parameters. All experiments were conducted using Gurobi 12.0 with a time limit of 60 seconds, MIP gap tolerance of 1%, and 4 parallel threads. For these experiments, we set $|\mathcal{I}| = 7$ tasks, $T = 30$ days, and $|\mathcal{S}| = 4$ scenarios, though the model accommodates arbitrary parameter configurations.

4.1. Experimental Design

We constructed six experimental scenarios varying the initial completion state g_0, total required hours G^{total}, and planning horizon T. Table 3 summarizes the experimental conditions.

Table 3: Experimental Scenarios and Results

Scenario	g_0	G^{total}	T	z^*
Early Stage	0.10	100	30	622.52
Baseline	0.30	100	30	480.63
Mid Stage	0.50	100	30	448.72
Late Stage	0.80	100	30	449.39
High Workload	0.30	150	30	480.63
Short Horizon	0.30	100	14	477.94

4.2. Main Results

The computational experiments reveal striking sensitivity of optimal allocations to the initial completion state. Figure 1 displays the relationship between initial completion rate and optimal objective value, showing a pronounced decrease from $z^* = 622.52$ at $g_0 = 0.1$ to $z^* = 448.72$ at $g_0 = 0.5$, after which the objective stabilizes.

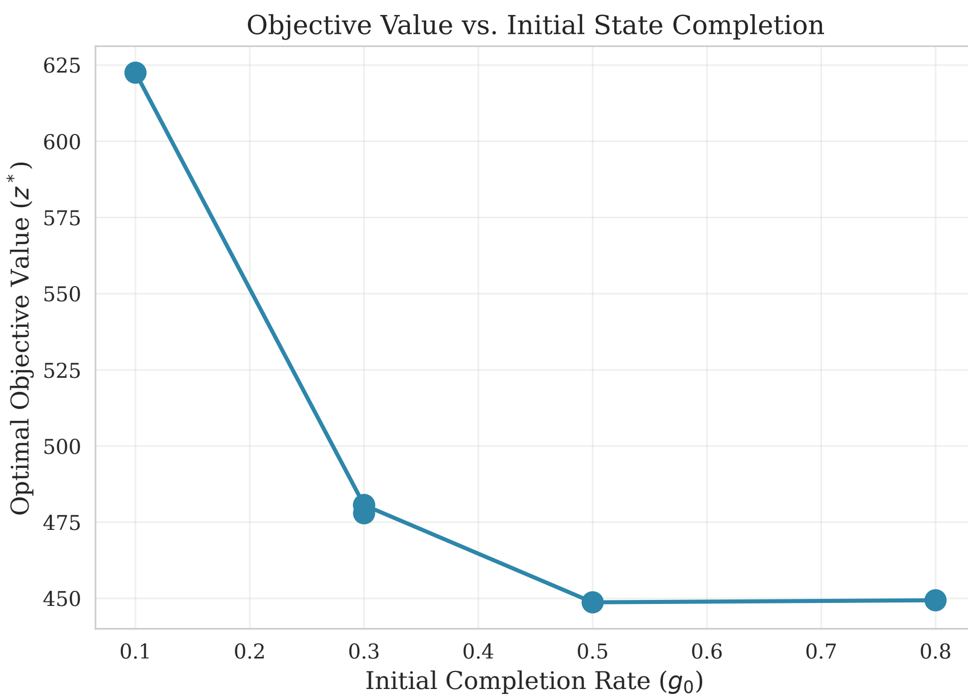

Figure 1: Optimal Objective Value as a Function of Initial Completion State

Table 4 presents the weekly hour allocations across task categories for each scenario. The results demonstrate substantial reallocation of resources as initial state varies.

4.3. Sensitivity Analysis

We computed sensitivity coefficients to quantify the marginal effect of initial state on optimal objective value. The sensitivity $\partial z^*/\partial g_0$ varies dramatically across the state space, measuring -709.45 at $g_0 = 0.1$ but diminishing to -94.84 at $g_0 = 0.5$ and approaching zero at $g_0 = 0.8$. This nonlinear sensitivity pattern indicates that early-stage decisions have outsized importance for overall value creation.

Table 4: Weekly Task Allocations by Scenario (Hours)

Scenario	g_0	Task 1	Task 2	Task 3
Early Stage	0.10	42.00	7.00	0.00
Baseline	0.30	31.35	22.66	0.00
Mid Stage	0.50	9.99	28.00	18.80
Late Stage	0.80	10.00	28.00	27.00

The correlation between initial completion rate and Task 1 allocation reaches $r = -0.91$, representing a strong negative relationship. Figure 2 illustrates the task allocation trajectories across state conditions, clearly showing the substitution pattern between Task 1 and Task 2.

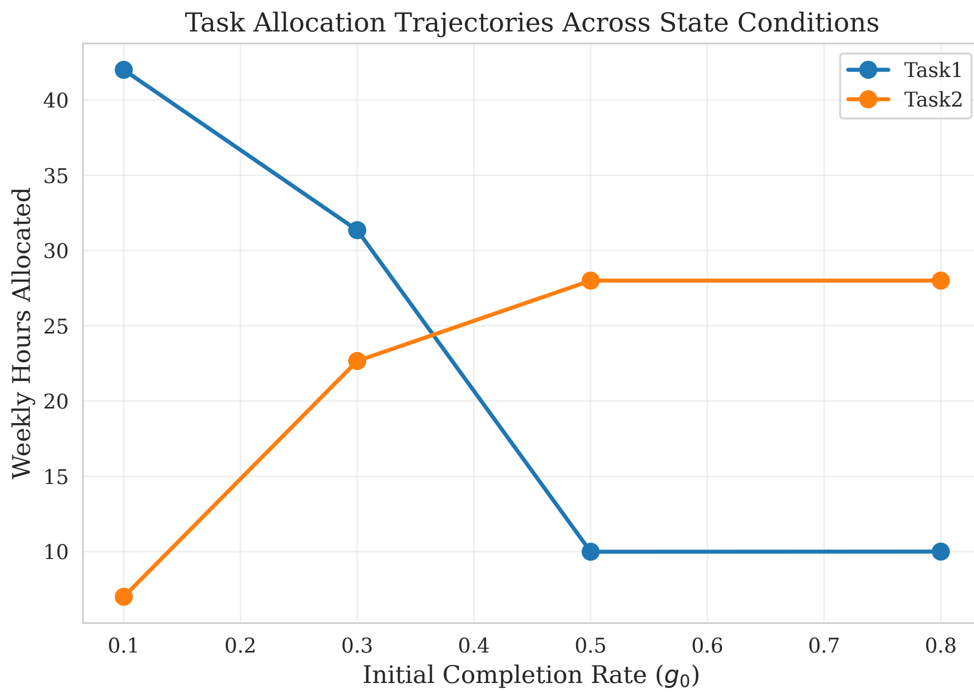

Figure 2: Task Allocation Trajectories Across Initial State Conditions

4.4. *Effect Size Analysis*

To assess the practical significance of the state-dependency, we computed Cohen's d effect size comparing early-stage ($g_0 = 0.1$) and late-stage ($g_0 = 0.8$) objective values. The effect size of $d = 2.0$ indicates a large practical difference, confirming that initial conditions substantially influence optimal outcomes.

The marginal value of planning horizon extension was computed by comparing 14-day and 30-day horizons. Each additional planning day yields approximately 0.17 utility units, suggesting modest but positive returns to extended planning.

5. Discussion

The computational results provide several insights with implications for both theory and practice.

5.1. *State-Dependent Priority Shifts*

The strong negative correlation between initial completion state and priority task allocation ($r = -0.91$) demonstrates that optimal resource allocation is fundamentally state-dependent. As progress accumulates on the critical task, the model rationally reallocates resources toward career development and scheduled obligations. This finding validates the rolling-horizon approach, which enables continuous re-optimization based on updated state information.

The observed substitution pattern aligns with intuitive notions of priority management: when urgent matters are well-controlled, attention shifts to important-but-not-urgent activities.

5.2. *Early-Stage Decision Importance*

The sensitivity analysis reveals that decisions made at low completion states have substantially larger marginal effects than those made at higher completion states. This asymmetry suggests that decision support systems should provide particularly careful guidance during early project phases, when the stakes for optimal allocation are highest.

5.3. *Practical Implications*

For practitioners designing personal productivity systems or resource allocation tools, these findings suggest several design principles. First, allocation recommendations should be conditioned on current system state rather than following static rules. Second, particular attention should be devoted to early-stage guidance, where suboptimal decisions carry the largest opportunity costs. Third, planning horizons of 30 days appear sufficient to capture relevant dynamics, with limited incremental value from longer horizons.

5.4. *Limitations*

Several limitations warrant acknowledgment. The model assumes known task parameters and scenario probabilities, whereas practical applications may face parameter uncertainty. The discrete scenario representation of recovery requirements represents a simplification of continuous uncertainty. Finally, the single-agent formulation does not capture potential coordination requirements in team settings.

6. Conclusion

This paper has developed and analyzed a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model captures state-dependent prioritization dynamics and uncertain recovery requirements within a computationally tractable framework.

Our computational experiments demonstrate that optimal allocation strategies exhibit strong sensitivity to initial conditions, with early-stage decisions having outsized importance for overall value creation. The observed resource substitution patterns between priority tasks and longer-term activities are consistent with rational priority management principles.

Several directions merit future investigation. Extensions to multi-stage stochastic programming with recourse decisions would enable more sophisticated uncertainty handling. Incorporation of learning effects and productivity dynamics could improve the model's descriptive accuracy. Field validation studies would establish the practical value of model-based recommendations.

Acknowledgments

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